

● HARD

● ALGEBRA

● ANSWER KEY

Linear equations in two variables

02

10 answers · Rationale-rich · Teach from doc

Q1**C**

C. $18x + Ry = 36$

Choice C is correct. The equation representing the linear relationship shown can be written in slope-intercept form $y = mx + b$, where m is the slope and b is the y -intercept of the line. The line shown passes through the points $(0, 6)$ and $(2, 0)$. Given two points on a line, x_1, y_1 and x_2, y_2 , the slope of the line can be calculated using the equation $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting $(0, 6)$ and $(2, 0)$ for x_1, y_1 and x_2, y_2 , respectively, in this equation yields $m = \frac{0 - 6}{2 - 0}$, which is equivalent to $m = -\frac{6}{2}$, or $m = -3$. Since $(0, 6)$ is the y -intercept, it follows that $b = 6$. Substituting -3 for m and 6 for b in the equation $y = mx + b$ yields $y = -3x + 6$. Adding $3x$ to both sides of this equation yields $3x + y = 6$. Multiplying this equation by 6 yields $18x + 6y = 36$. It follows that the equation $18x + Ry = 36$, where R is a positive constant, represents this relationship.

Choice A is incorrect. The graph of this relationship passes through the point $(0, 2)$, not $(0, 6)$.

Choice B is incorrect. The graph of this relationship passes through the point $(0, 2)$, not $(0, 6)$.

Choice D is incorrect. The graph of this relationship passes through the point $(-2, 0)$, not $(2, 0)$.

Q2**5**

The correct answer is 5. It's given that the equation $10x + 15y = 85$ represents the situation, where x is the number of on-site training courses, y is the number of online training courses, and 85 is the total number of hours of training courses the apprentice has enrolled in.

Therefore, $10x$ represents the number of hours the apprentice has enrolled in on-site training courses, and $15y$ represents the number of hours the apprentice has enrolled in online training courses. Since x is the number of on-site training courses and y is the number of online training courses the apprentice has enrolled in, 10 is the number of hours each on-site course takes and 15 is the number of hours each online course takes. Subtracting these numbers gives $15 - 10$, or 5 more hours each online training course takes than each on-site training course.

Q3**1.2**

Accept also: 6/5

The correct answer is $\frac{6}{5}$. It's given that line j is perpendicular to line k in the xy -plane. This means that the slope of line j is the opposite reciprocal of the slope of line k . For a line that passes through the points x_1, y_1 and x_2, y_2 in the xy -plane, the slope of the line can be calculated as $\frac{y_2 - y_1}{x_2 - x_1}$. It's shown that line k passes through the points $(-6, 0)$ and $(0, -5)$ in the xy -plane. Substituting -6 for x_1 , 0 for y_1 , 0 for x_2 , and -5 for y_2 in $\frac{y_2 - y_1}{x_2 - x_1}$ yields $\frac{-5 - 0}{0 - (-6)}$, or $-\frac{5}{6}$. The opposite reciprocal of $-\frac{5}{6}$ is $\frac{6}{5}$. Therefore, the slope of line j is $\frac{6}{5}$. Note that $6/5$ and 1.2 are examples of ways to enter a correct answer.

Q4**B****B.** $3x + sy = 18s$

Choice B is correct. The linear relationship between x and y can be represented by an equation of the form $y - y_1 = mx - x_1$, where m is the slope of the graph of the equation in the xy -plane and x_1, y_1 is a point on the graph. The slope of a line can be found using two points on the line and the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Each value of x and its corresponding value of y in the table can be represented by a point x, y . Substituting the points $-s, 21$ and $s, 15$ for x_1, y_1 and x_2, y_2 , respectively, in the slope formula yields $m = \frac{15 - 21}{s - (-s)}$, which gives $m = \frac{-6}{2s}$, or $m = -\frac{3}{s}$. Substituting $-\frac{3}{s}$ for m and the point $s, 15$ for x_1, y_1 in the equation $y - y_1 = mx - x_1$ yields $y - 15 = -\frac{3}{s}x - s$. Distributing $-\frac{3}{s}$ on the right-hand side of this equation yields $y - 15 = -\frac{3x}{s} + 3$. Adding 15 to each side of this equation yields $y = -\frac{3x}{s} + 18$. Multiplying each side of this equation by s yields $sy = -3x + 18s$. Adding $3x$ to each side of this equation yields $3x + sy = 18s$. Therefore, the equation $3x + sy = 18s$ represents this relationship.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q5**D****D.** $\frac{7}{2}$

Choice D is correct. The x -coordinate a of the x -intercept $a, 0$ can be found by substituting 0 for y in the given equation, which gives $7x + 20 = -31$, or $7x = -31$. Dividing both sides of this equation by 7 yields $x = -\frac{31}{7}$. Therefore, the value of a is $-\frac{31}{7}$. The y -coordinate b of the y -intercept $0, b$ can be found by substituting 0 for x in the given equation, which gives $7(0) + 2y = -31$, or $2y = -31$. Dividing both sides of this equation by 2 yields $y = -\frac{31}{2}$. Therefore, the value of b is $-\frac{31}{2}$. It follows that the value of $\frac{b}{a}$ is $\frac{-\frac{31}{2}}{-\frac{31}{7}}$, which is equivalent to $\frac{31}{2} \cdot \frac{7}{31}$, or $\frac{7}{2}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q6**A****A.** 0

Choice A is correct. It is given that line ℓ is perpendicular to a line whose equation is $x = 2$. A line whose equation is a constant value of x is vertical, so ℓ must therefore be horizontal. Horizontal lines have a slope of 0, so ℓ has a slope of 0.

Choice B is incorrect. A line with slope $-\frac{1}{2}$ is perpendicular to a line with slope 2. However, the line with equation $x = 2$ is vertical and has undefined slope (not slope of 2). Choice C is incorrect. A line with slope -2 is perpendicular to a line with slope $\frac{1}{2}$. However, the line with equation $x = 2$ has undefined slope (not slope of $\frac{1}{2}$). Choice D is incorrect; this is the slope of the line $x = 2$ itself, not the slope of a line perpendicular to it.

Q7**-10**

The correct answer is -10. A line in the xy -plane can be represented by the equation $y = mx + b$, where m is the slope of the line and b is the y -coordinate of the y -intercept. It's given that line p has a slope of $-\frac{5}{3}$. Therefore, $m = -\frac{5}{3}$. It's also given that line p has an x -intercept of $-6, 0$. Therefore, when $x = -6, y = 0$. Substituting $-\frac{5}{3}$ for m , -6 for x , and 0 for y in the equation $y = mx + b$ yields $0 = -\frac{5}{3}(-6) + b$, which is equivalent to $0 = 10 + b$. Subtracting 10 from both sides of this equation yields $-10 = b$. Therefore, the y -coordinate of the y -intercept of line p is -10 .

Q8**D**

D. $-\frac{17}{6}, 0$

Choice D is correct. The equation of line h can be written in slope-intercept form $y = mx + b$, where m is the slope of the line and b is the y -intercept of the line. It's given that line h contains the points $18, 130, 23, 160$, and $26, 178$. Therefore, its slope m can be found as $\frac{160 - 130}{23 - 18}$, or 6 . Substituting 6 for m in the equation $y = mx + b$ yields $y = 6x + b$. Substituting 130 for y and 18 for x in this equation yields $130 = 6(18) + b$, or $130 = 108 + b$. Subtracting 108 from both sides of this equation yields $22 = b$. Substituting 22 for b in $y = 6x + b$ yields $y = 6x + 22$. Since line k is the result of translating line h down 5 units, an equation of line k is $y = 6x + 22 - 5$, or $y = 6x + 17$. Substituting 0 for y in this equation yields $0 = 6x + 17$. Solving this equation for x yields $x = -\frac{17}{6}$. Therefore, the x -intercept of line k is $-\frac{17}{6}, 0$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q9**C**

C. $\frac{24}{7}$

Choice C is correct. It's given from the graph that the points 0, 7 and 8, 0 lie on the line. For two points on a line, x_1, y_1 and x_2, y_2 , the slope of the line can be calculated using the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting 0, 7 for x_1, y_1 and 8, 0 for x_2, y_2 in this formula, the slope of the line can be calculated as $m = \frac{0 - 7}{8 - 0}$, or $m = -\frac{7}{8}$. It's also given that the point $d, 4$ lies on the line.

Substituting $d, 4$ for x_1, y_1 , 8, 0 for x_2, y_2 , and $-\frac{7}{8}$ for m in the slope formula yields $-\frac{7}{8} = \frac{0 - 4}{8 - d}$, or $-\frac{7}{8} = \frac{-4}{8 - d}$. Multiplying both sides of this equation by $8 - d$ yields $-\frac{7}{8}(8 - d) = -4$. Expanding the left-hand side of this equation yields $-7 + \frac{7}{8}d = -4$. Adding 7 to both sides of this equation yields $\frac{7}{8}d = 3$. Multiplying both sides of this equation by $\frac{8}{7}$ yields $d = \frac{24}{7}$. Thus, the value of d is $\frac{24}{7}$.

Choice A is incorrect. This is the value of y when $x = 4$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q10**189/5**

Accept also: 37.8

The correct answer is $\frac{189}{5}$. A y -intercept of a graph in the xy -plane is a point where the graph intersects the y -axis, which is a point with an x -coordinate of 0. Substituting 0 for x in the given equation yields $\frac{30}{7} = -\frac{5y}{9} + 21$, or $0 = -\frac{5y}{9} + 21$. Subtracting 21 from both sides of this equation yields $-21 = -\frac{5y}{9}$. Multiplying both sides of this equation by -9 yields $189 = 5y$. Dividing both sides of this equation by 5 yields $\frac{189}{5} = y$. Therefore, the y -coordinate of the y -intercept of the graph of the given equation in the xy -plane is $\frac{189}{5}$. Note that 189/5 and 37.8 are examples of ways to enter a correct answer.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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